# Post-Incident Report: SNMPv3 Polling Failures (#2001–#2003)

NOC-Monitoring-Lab · Prepared by Randy · October 29, 2026

## Summary

Three routers in the NOC-Monitoring-Lab (VyOS-1, VyOS-2 and a MikroTik CHR) stopped being polled by LibreNMS after three separate SNMPv3 configuration changes. Ping and OSPF stayed healthy on every router the whole time, which isolated the fault to the SNMP management plane. Each incident was diagnosed from its symptoms, traced to a root cause, and fixed. Polling was fully restored by 15:11.

## Environment

- Monitoring server: LibreNMS on an Ubuntu Server VM (10.0.100.10)
- Devices: VyOS-1 (1.1.1.1), VyOS-2 (2.2.2.2), MikroTik CHR 7.24.5 (3.3.3.3), connected in an OSPF triangle
- SNMP: v3 authPriv (SHA authentication, AES privacy), polled over each router's loopback
- Before the incidents, SNMPv2c was removed from all three routers and verified: v2c queries time out, v3 queries return the sysName of each router

## Timeline (October 29, 2026, LibreNMS eventlog)

| Time | Device | Event |
|---|---|---|
| 12:55 | 3.3.3.3 | Reboot (uptime 271 s), unrelated to the incidents |
| 13:55:03 | 1.1.1.1 | Down, from snmp check |
| 13:55:09 | 3.3.3.3 | Down, from snmp check |
| 14:05:04 | 2.2.2.2 | Down, from snmp check |
| 15:08:01 | 1.1.1.1 | Up, from snmp check |
| 15:09:16 | 3.3.3.3 | Up, from snmp check |

| Time | Device | Event |
|---|---|---|
| 15:11:42 | 2.2.2.2 | Up, from snmp check |

Every state change came from the SNMP check and none from ICMP, which is the signature of an SNMP-only fault.

## Incident #2001: VyOS-1 down in LibreNMS (easy)

- **Symptom:** LibreNMS showed vyos-1 down. Ping to 1.1.1.1 and OSPF were fine, and no change had been recorded.
- **Layer:** User authentication. Reachability was proven fine, so the fault was in the SNMPv3 credentials.
- **Evidence:** `compare saved` showed the stored auth encrypted-password differed from the saved baseline. The `show configuration` output alone could not reveal this, because a hash is not readable.
- **Root cause:** The SNMPv3 user's authentication password had been changed.
- **Fix:** Restored the known-good auth password and committed.
- **Verification:** `snmpget -v3` from the LibreNMS VM returned vyos-1.

## Incident #2002: CHR flagged SNMP unreachable (medium)

- **Symptom:** The CHR showed down or unreachable. Ping and OSPF were fine, and another engineer said the credentials had not changed.
- **Layer:** Privacy (encryption).
- **Evidence:** The CHR's own syslog, already forwarded to LibreNMS, recorded `changed by console:admin@serial0 (/snmp community set ... encryption-protocol=DES)`. `/snmp community print detail` showed DES on the `randy` entry, while LibreNMS and the other routers use AES.
- **Root cause:** The privacy protocol was changed from AES to DES. The credentials were unchanged, but the protocol was not, so the statement "the credentials haven't changed" was only partly true.
- **Fix:** Set the encryption protocol back to AES.
- **Verification:** `/snmp community print detail` shows authentication-protocol=SHA1 and encryption-protocol=AES again, and LibreNMS marked 3.3.3.3 up.

## Incident #2003: VyOS-2 stopped polling (hard)

- **Symptom:** vyos-2 stopped polling, while the SNMP user, auth and privacy lines looked unchanged and the passwords were correct.
- **Layer:** SNMP engine ID.
- **Evidence:** The engine ID had been changed from 000000000000000000000**02** to 00000099. It also broke the lab's engine ID standard (24 hex digits ending in the device number).
- **Root cause:** SNMPv3 keys are localized to the engine ID. After the engine ID changed, the same passwords produced different keys, so authentication failed even though every user line looked identical.
- **Fix:** Restored the original engine ID.
- **Verification:** `compare saved` showed no changes, and the auth and privacy hashes matched the originals exactly, which confirms the localization behavior. LibreNMS marked 2.2.2.2 up.

## Lessons Learned

1. Ping up and SNMP down points to the management plane. Check credentials, protocols and engine IDs, not routing.
2. The same symptom can have different causes. A wrong password and a changed engine ID can both produce an authentication failure, so the error text alone does not identify the layer.
3. Comparing against a saved baseline is more reliable than reading the config. Hashed passwords cannot be read back, but a diff shows that they changed.
4. Device syslog gave attribution. The CHR recorded who made the change and what the change was.
5. "The credentials haven't changed" needs verification. Protocols and engine IDs are part of the credential set.

## Follow-Up Actions

- Set up automated config backups with Oxidized, so baselines and diffs exist without relying on a manual save
- Document the engine ID template as a lab standard
- Add this report and screenshots to the project README